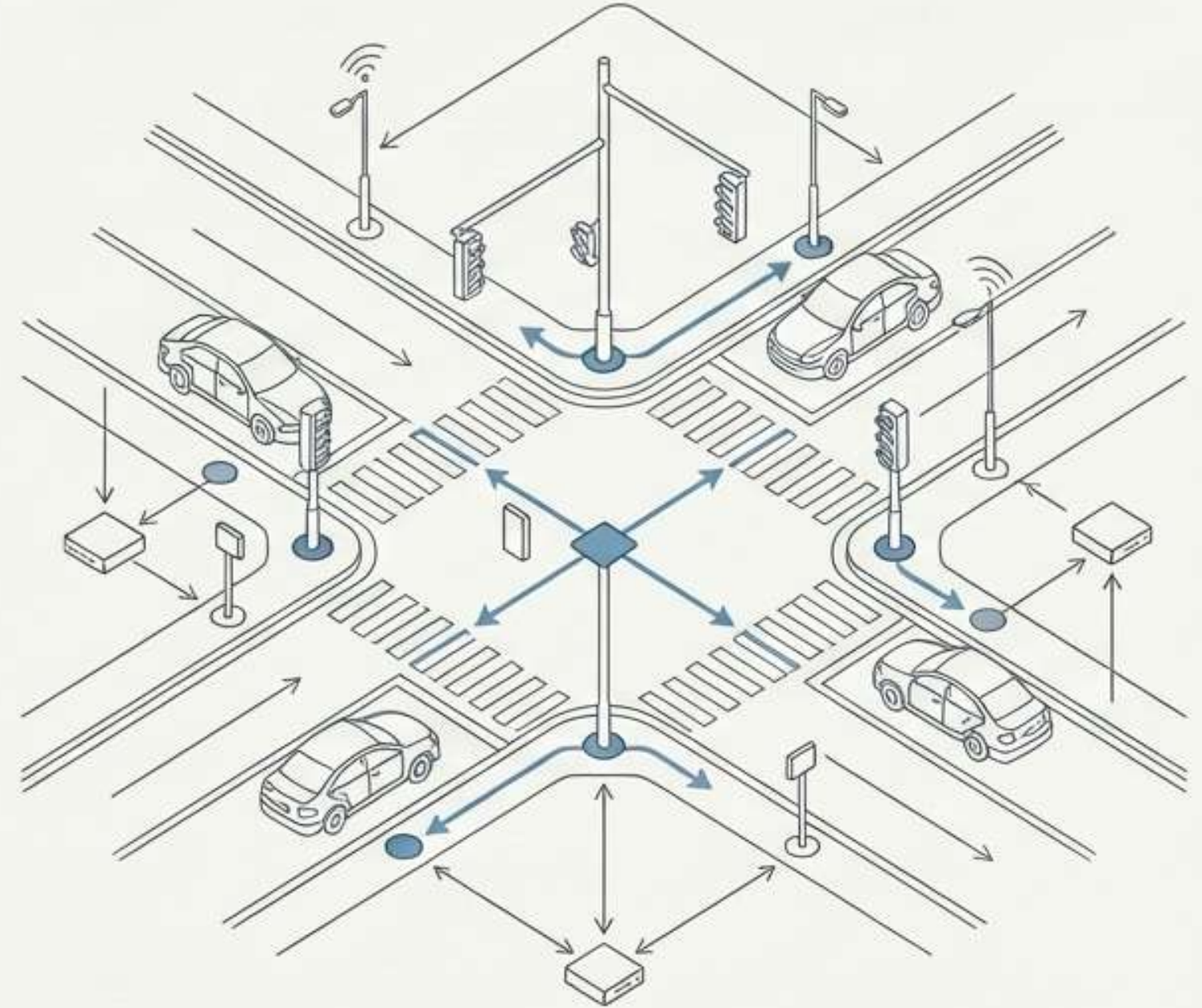


IoT-Based Smart Traffic System

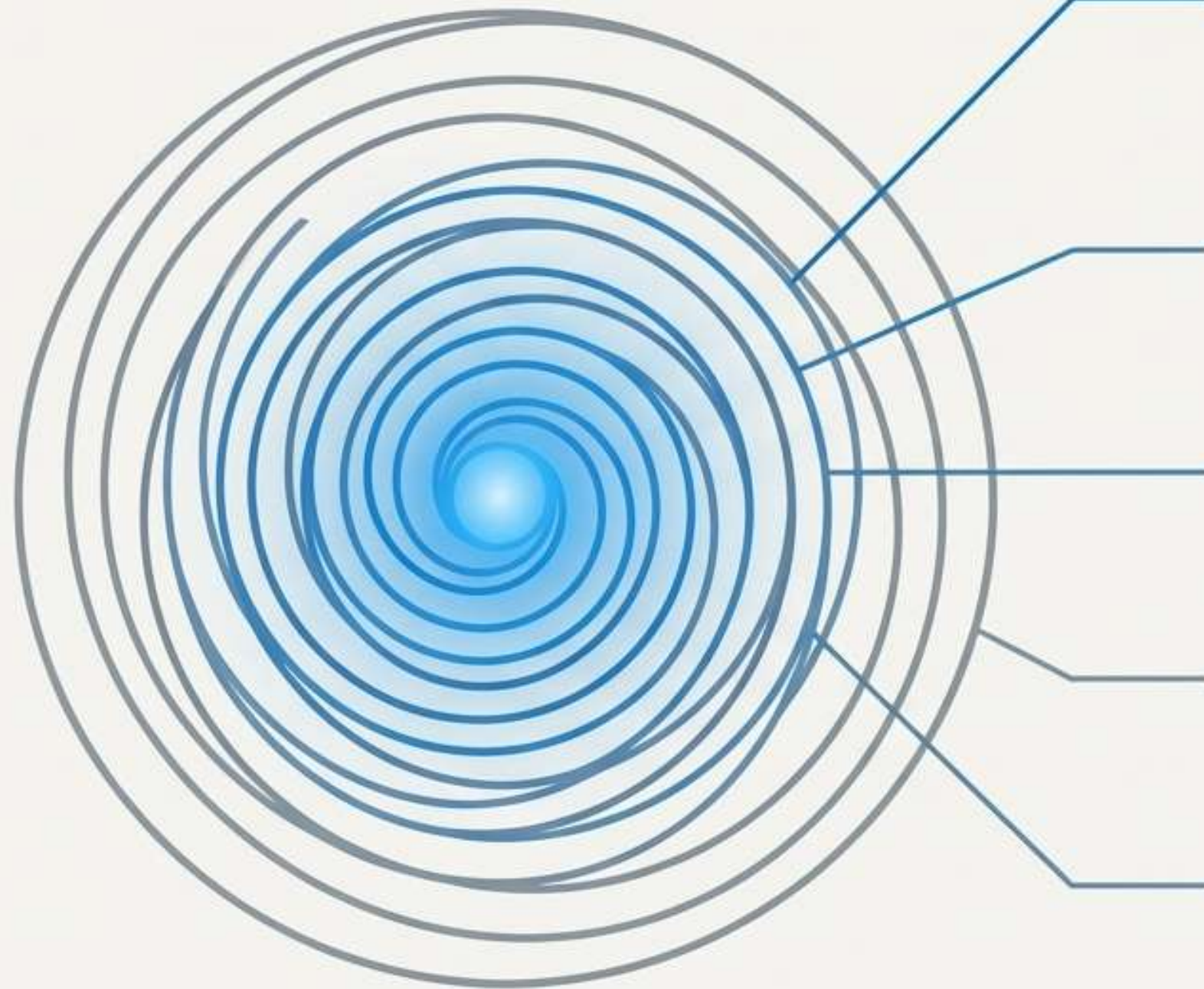
Intelligent, Real-Time Traffic Control & Predictive Analysis

What is IoT?

A network of physical devices—sensors, actuators, and microcontrollers—connected via the internet to collect, exchange, and act on real-world data.



Redefining Urban Mobility: Core Objectives



01. Real-Time Monitoring

Aggregate external traffic API streams (TomTom/OpenStreetMap) to capture live vehicle density.

02. Dynamic Signal Control

Auto-adjust timings to replace outdated fixed-timer systems.

03. Congestion Prediction

Utilize ML to forecast build-ups before they occur.

04. Cloud Dashboards

Stream data for remote monitoring and automated alerts.

05. Urban Impact

Reduce travel delays and lower vehicle fuel consumption.

The Cost of Stagnation

₹1.5L Cr

ANNUAL ECONOMIC LOSS IN INDIA DUE TO
URBAN TRAFFIC CONGESTION.

30%

URBAN FUEL CONSUMPTION ENTIRELY
WASTED IN STOP-AND-GO TRAFFIC.



3.7 hrs

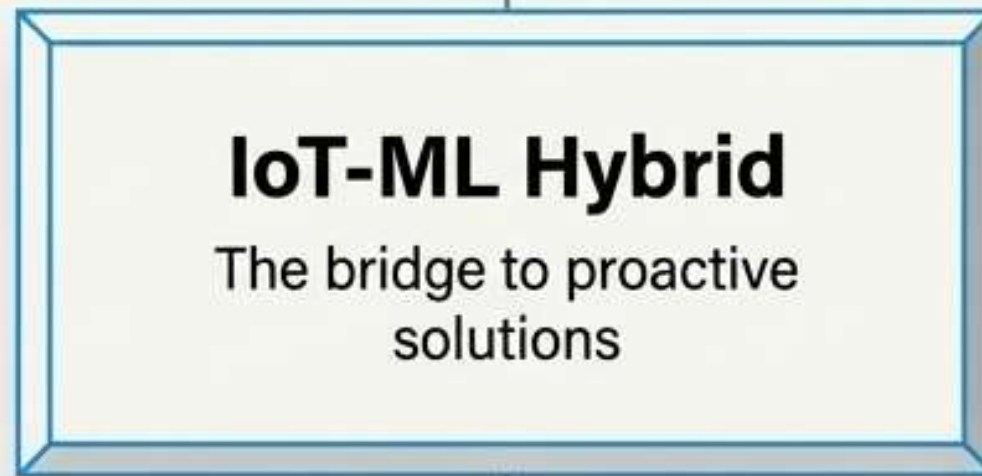
AVERAGE TIME WASTED PER DRIVER ANNUALLY.

Rapid urbanization has outpaced traditional fixed-timer signals, which cannot adapt to accidents, events, or sudden peak

The Bridge to Proactive Solutions

Status Quo

- Fixed Timing
(Rigid)
- Human Dependency
(Slow response)
- Information Silos
(No data sharing)

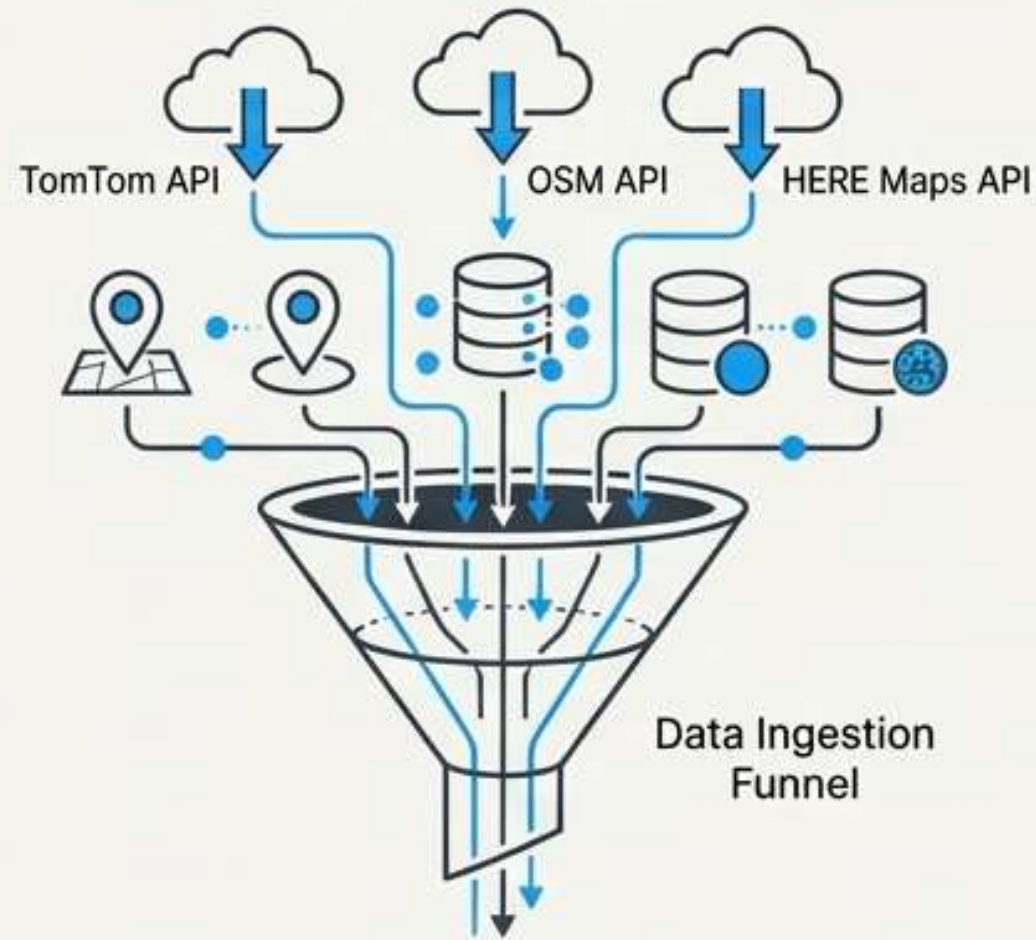


Change Enablers

- Economic Impact
(Optimizing flow)
- Environmental Degradation
(Reducing emissions)
- Emergency Services
(Dynamic routing)

Intelligent Observation & Visualization

Multi-Source Data Ingestion



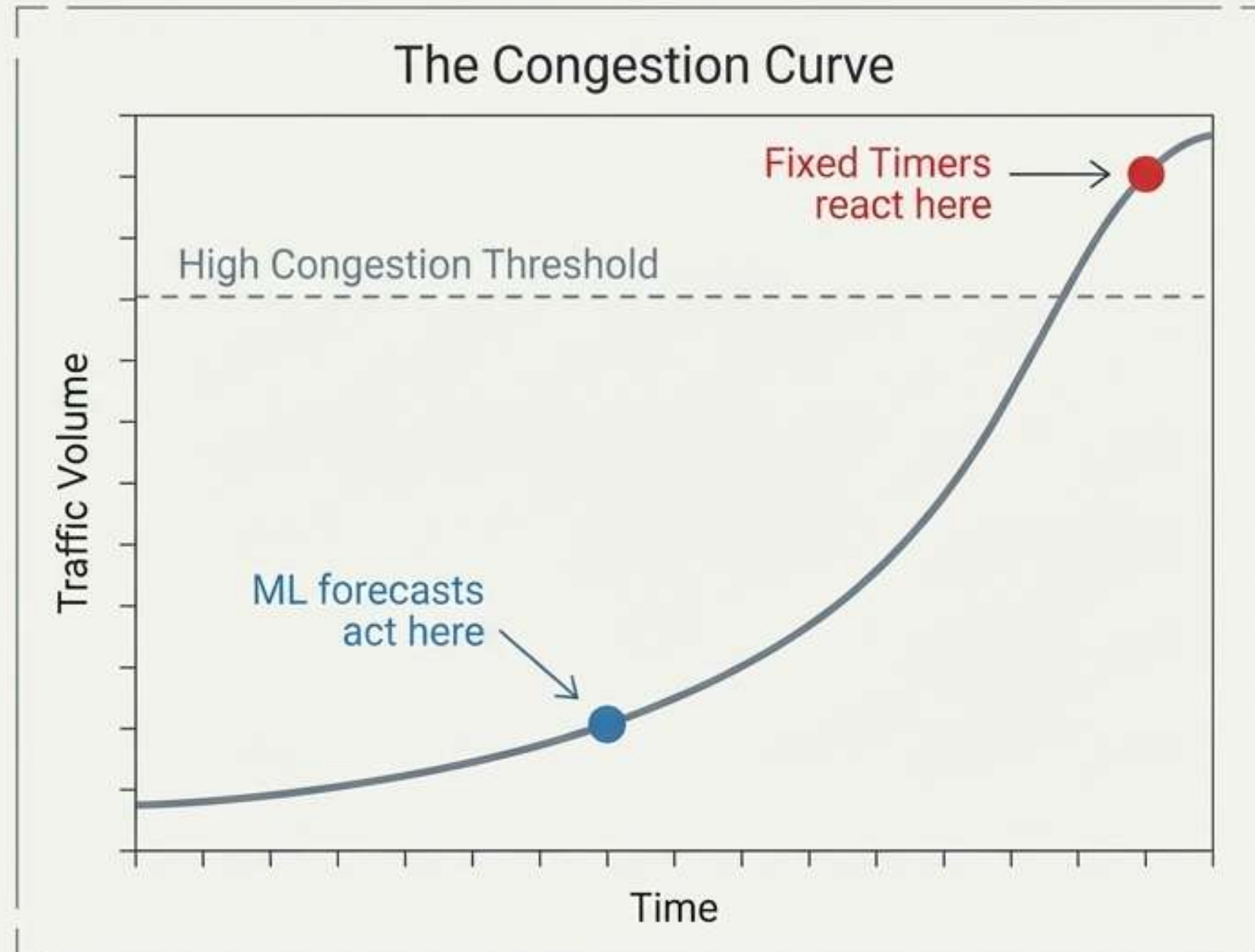
- Integrating TomTom, OSM, and HERE Maps APIs.
- Ingesting live vehicle velocity, travel times, and grid density.

Cloud-Based Interface



- AWS/Firebase storage for historical logging.
- Mobile apps and web dashboards providing real-time alerts, traffic status, and ETAs.

Proactive Forecasting over Reactive Management



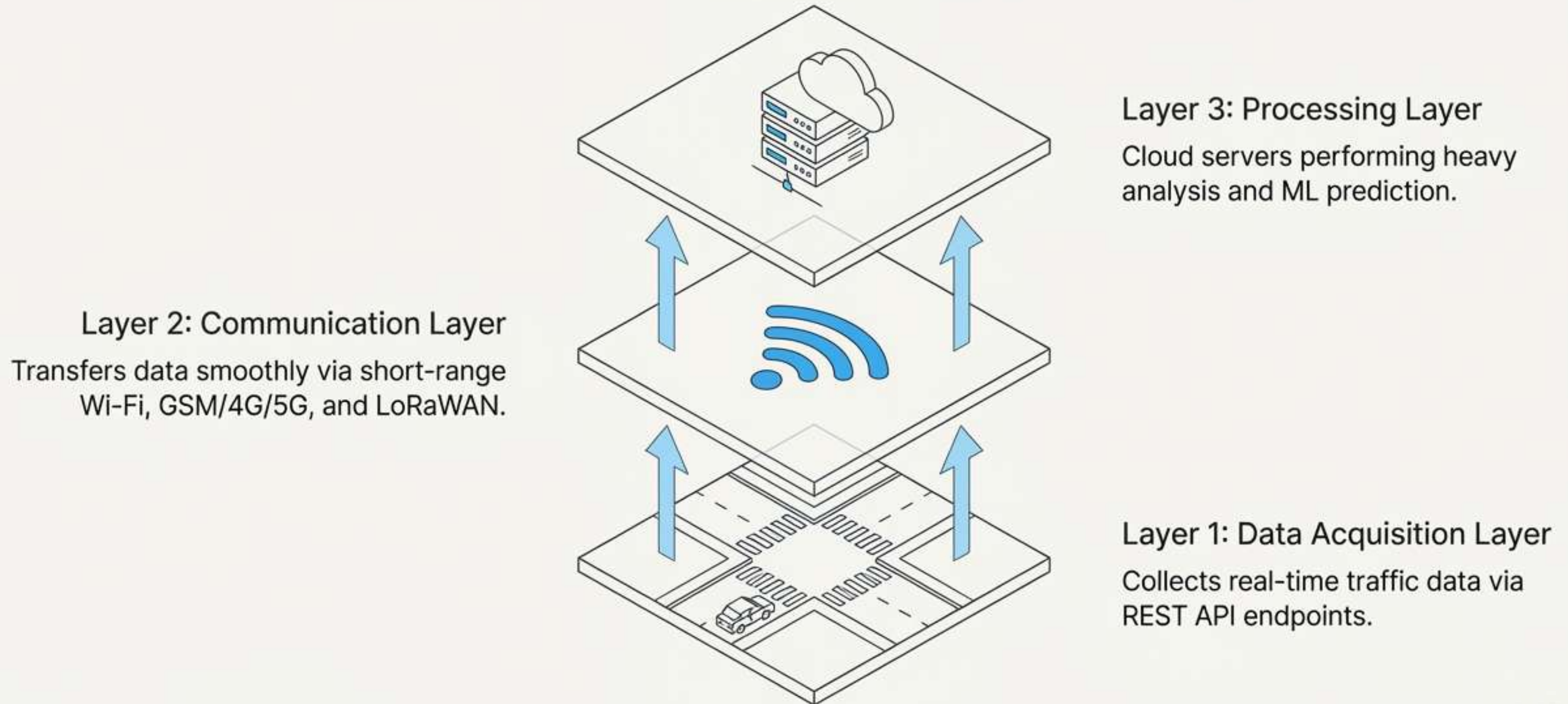
Smart Congestion Detection

Classifies live volume into Low, Medium, and High traffic using threshold-based analysis.

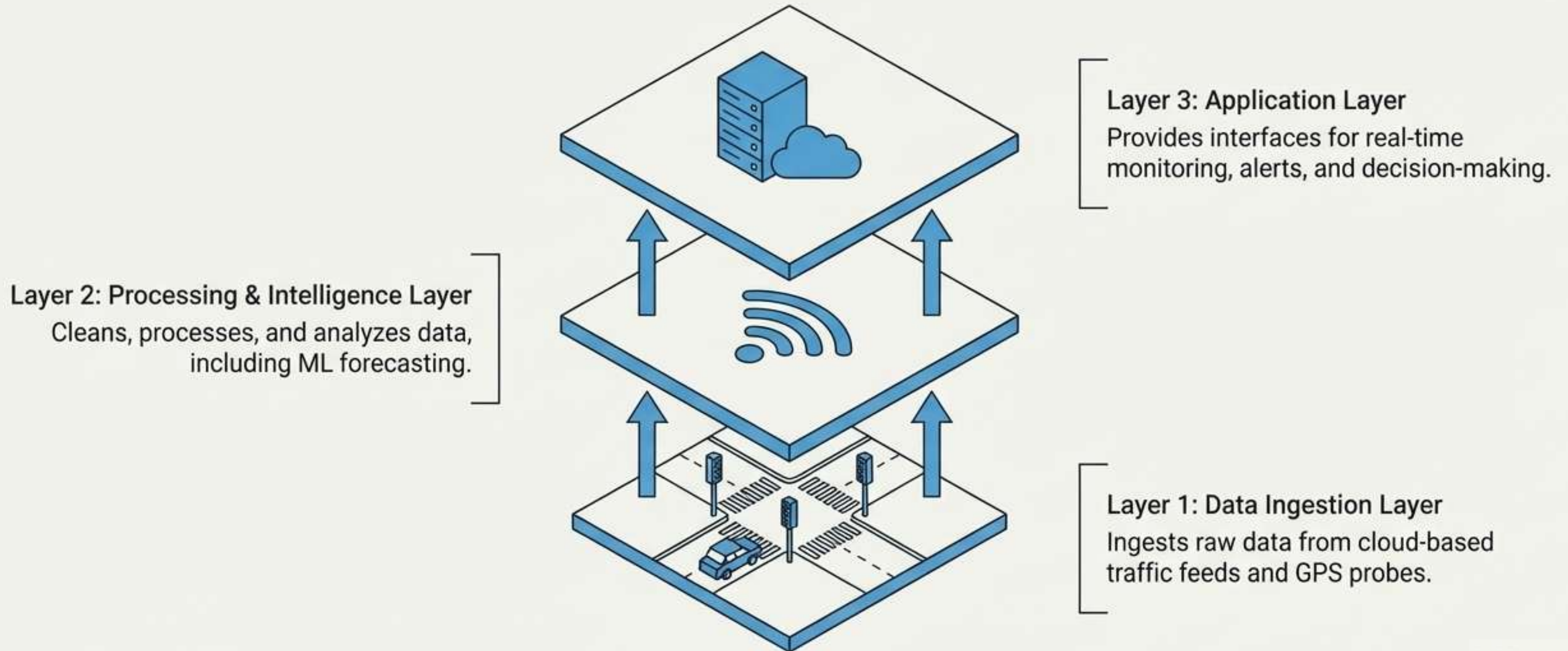
Machine Learning Forecasting

ARIMA (Time-series prediction) and LSTM (Deep learning-based prediction) anticipate future conditions.

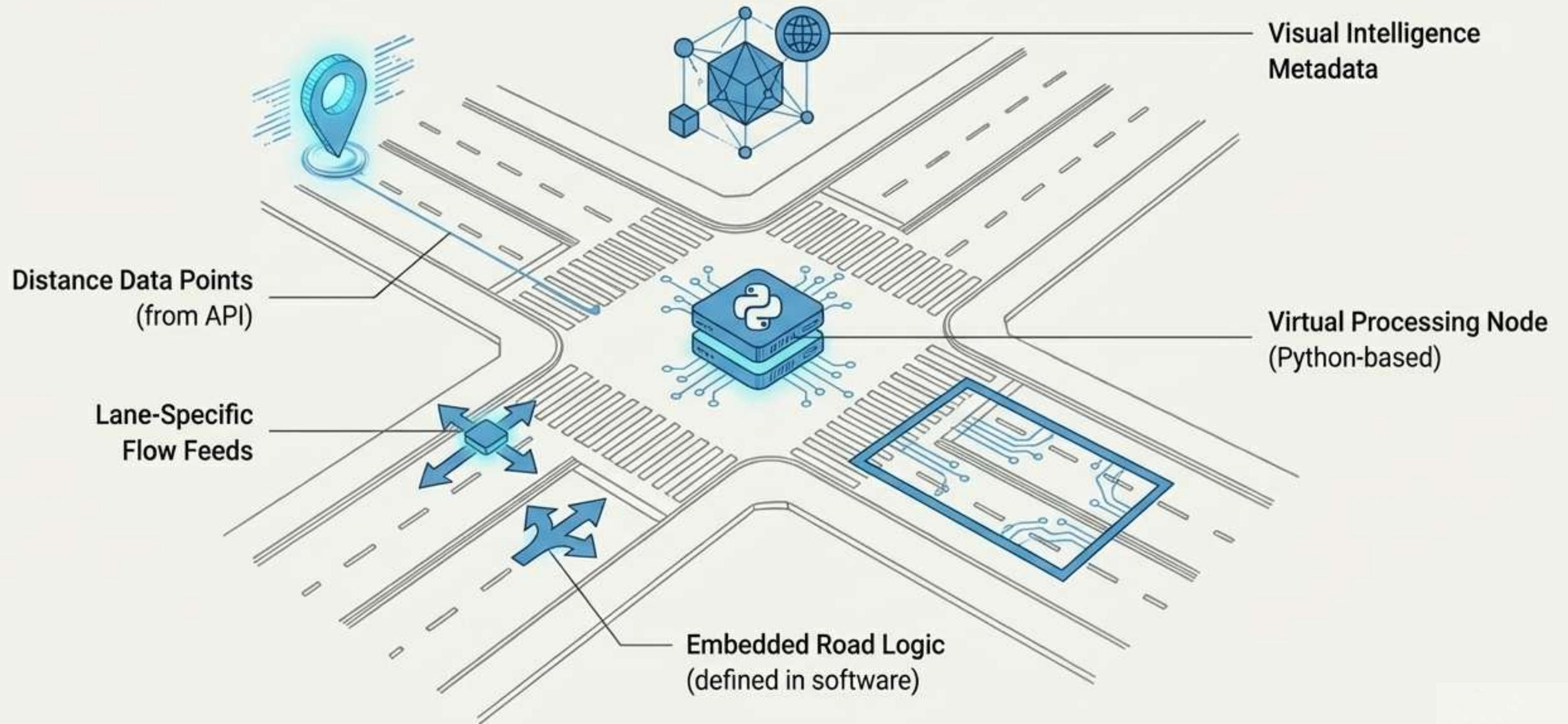
The Three-Layer Architecture (Network View)



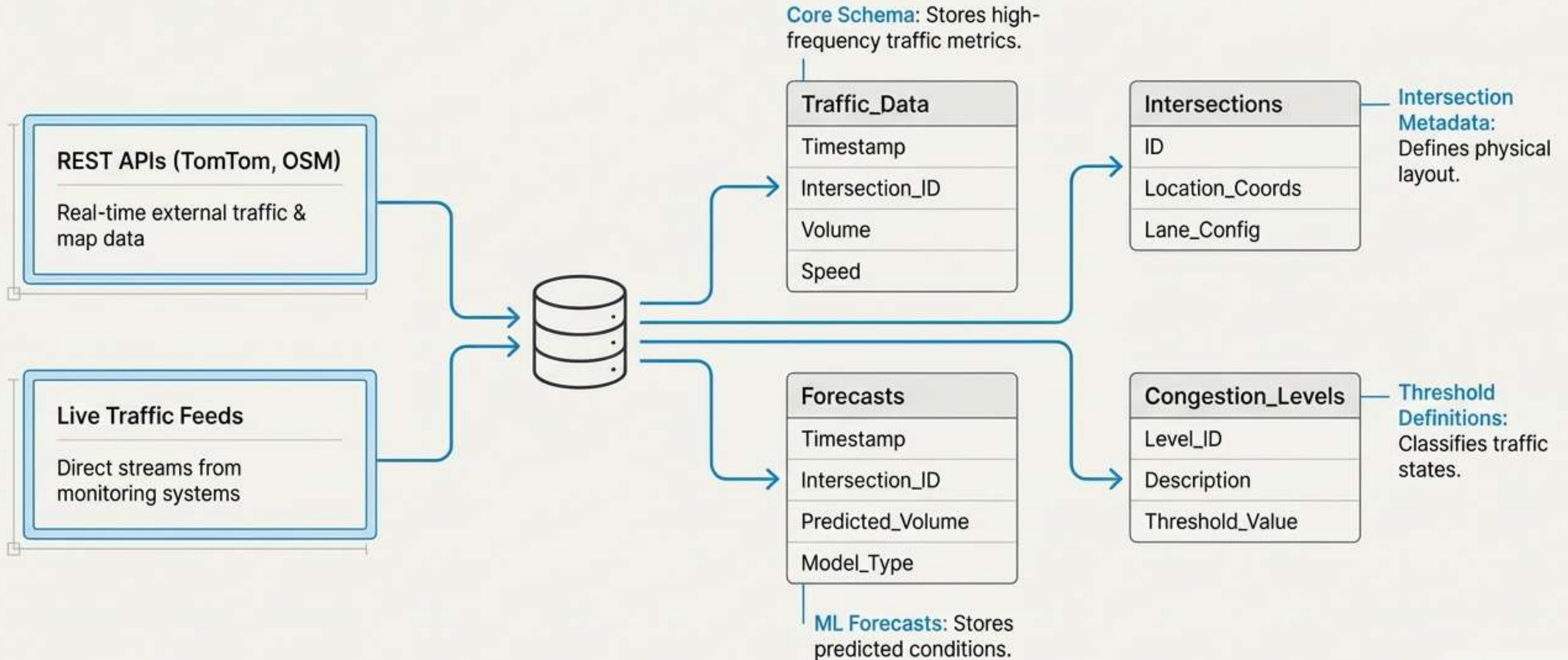
The Three-Layer Architecture (Data Intelligence View)



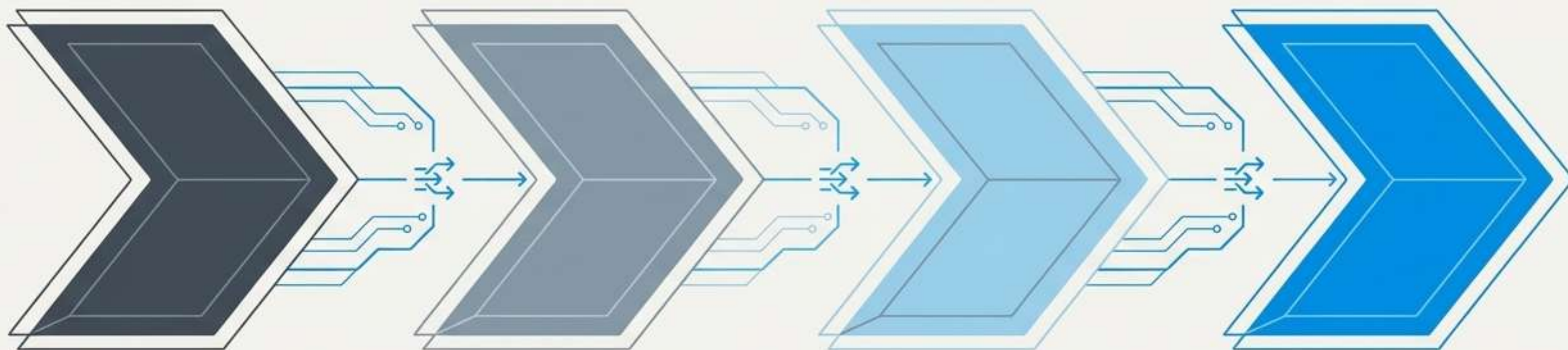
Virtual Intersection Mapping & Data Points



Database Schema & Data Source Design



The Data to Decision Pipeline



Step 1: Raw Collection

Ingesting high-frequency telemetry and diverse inputs from external traffic APIs.

Step 2: Processing & Cleaning

Removing noise, handling missing values, and extracting core features (speed, density, flow rate).

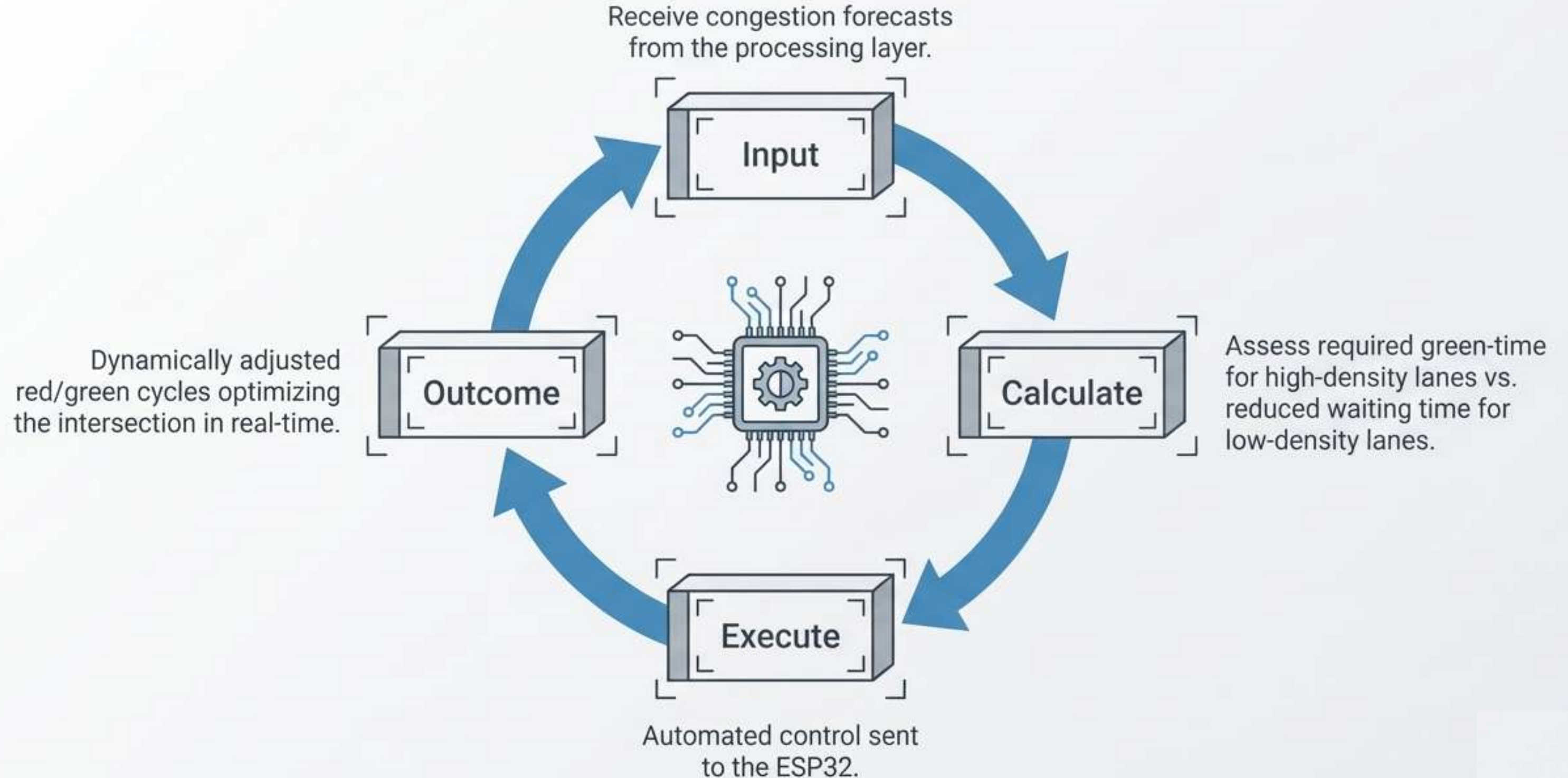
Step 3: Threshold Classification

Categorizing traffic states (Low flow vs. Medium delay vs. High congestion).

Step 4: ML Ingestion

Funneling structured data into Linear Regression, ARIMA, and LSTM models for forecasting.

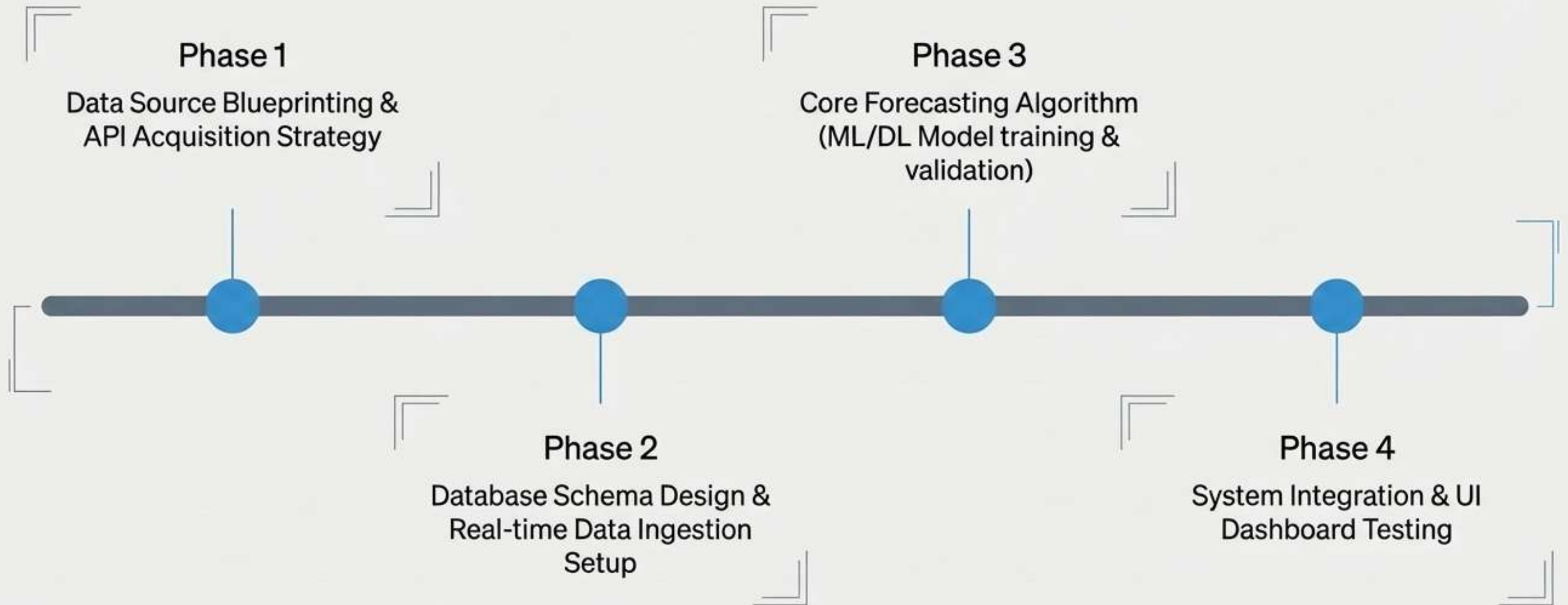
Algorithmic Signal Execution



Team Deployment Matrix

Vignesh	Introduction, Objectives & Problem Definition
Jennie	Motivation & Requirement Analysis
Ushasri	Feature Mapping & Functional Requirements
Ankur	System Architecture & Software Design
Minhaz	Algorithm Development & Methodology
Sathwika	Project Coordination & Progress Tracking
Archisha	Expected Outcomes, References & Snapshots

Project Coordination & Tracking



Implementation Status

Mission Control

Proactive Management

Forecasting models actively deployed to prevent bottlenecks.

Adaptive Timing

Microcontrollers making live density adjustments.

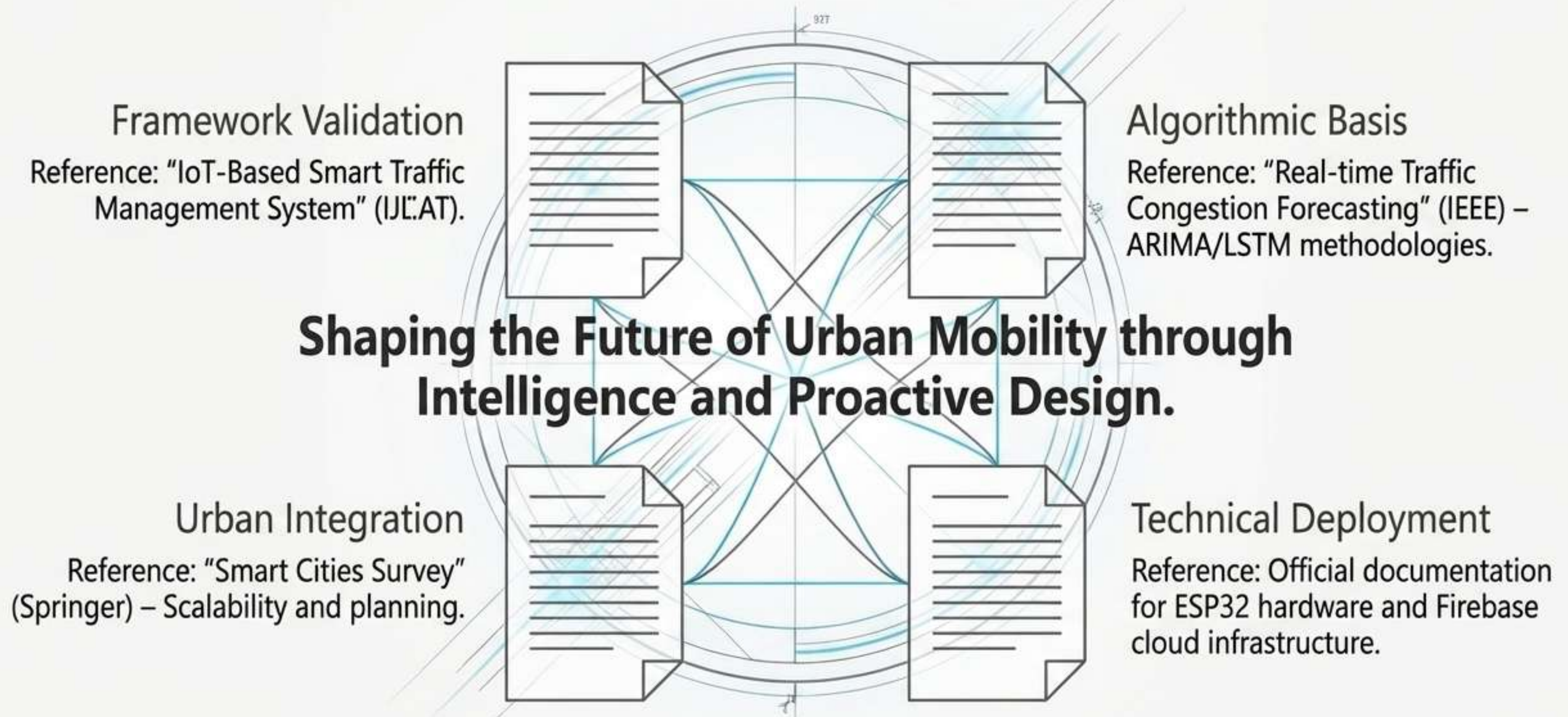
Public Awareness

Web-based dashboard framework operational.

Urban Insights

Logging historical data for long-term urban planning.

Research Foundation & The Future of Smart Cities



A large, faint circular graphic in the background. It features concentric circles and radial lines. Two specific points on the circle are highlighted with bright teal light and radiating lines, suggesting a focus or data points. The overall aesthetic is clean, modern, and technological.

Thank You.

Shaping the Future of Smart Cities through Intelligence and Proactive Design.